

Week 4- Implementing Cloud Logging and Monitoring

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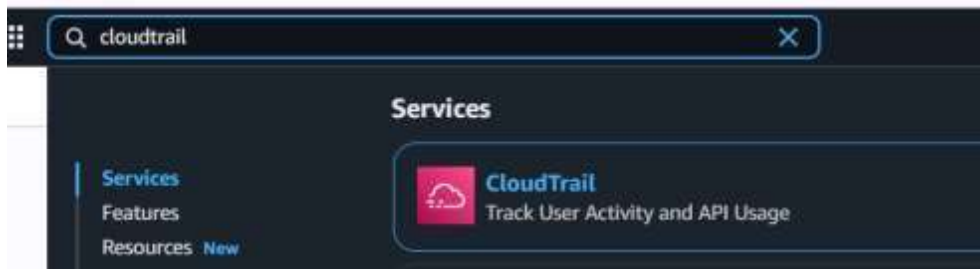
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- ❖ **Task-**
- 1) Enable CloudTrail/Cloud Audit Logs.
 - 2) Configure alerts for suspicious activity.
 - 3) Document findings.

Step Enable Cloud Trail/ Cloud Audit Logs.

Step 1: Login to AWS Console

Step 2: Search and open CloudTrail from the Services menu.



Step 3: Click Create trail.



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Step 4- Fill in trail Name cloudtrail-1, and then click create trail.

The screenshot shows the 'Quick trail create' page in the AWS CloudTrail console. The 'Trail name' field is filled with 'cloudtrail-1'. The 'Trail log bucket and folder' field shows 'aws-cloudtrail-logs-355433852359-b5d20482'. A note states: 'Though there is no cost to log these events, you incur charges for the S3 bucket that we create to store your logs.' At the bottom, there are 'Cancel' and 'Create trail' buttons.

Trail details
Start logging management events by creating a trail with simplified settings. Logs are sent to an S3 bucket we create on your behalf. To choose a different bucket or additional events, go to the full [Create trail workflow](#).

A trail created in the console is a multi-region trail. [Learn more](#)

Trail name
Enter a display name for your trail.

3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.

Trail log bucket and folder

Logs will be stored in `aws-cloudtrail-logs-355433852359-b5d20482/AWSLogs/355433852359`

Buttons: Cancel, Create trail

The screenshot shows the 'Trails' page in the AWS CloudTrail console. It displays a table with two trails: 'cloudtrail-1' and 'management-events'. Both trails are in the 'Europe (Stockholm)' region, are multi-region trails, and are currently 'Disabled'. The 'cloudtrail-1' trail logs to the S3 bucket 'aws-cloudtrail-logs-355433852359-b5d20482'. The 'management-events' trail logs to the S3 bucket 'aws-cloudtrail-logs-355433852359-b5d20482'. Both trails have a status of 'Logging'.

Name	Home region	Multi-region trail	ARN	Insights	Organization trail	S3 bucket	Log file prefix	CloudWatch Logs log group	Status
cloudtrail-1	Europe (Stockholm)	Yes	arn:aws:cloudtrail:eu-north-1:355433852359:trail/cloudtrail-1	Disabled	No	aws-cloudtrail-logs-355433852359-b5d20482			Logging
management-events	Europe (Stockholm)	Yes	arn:aws:cloudtrail:eu-north-1:355433852359:trail/management-events	Disabled	No	aws-cloudtrail-logs-355433852359-b5d20482			Logging

Step 5- Set up S3 Bucket to store logs: create new bucket.

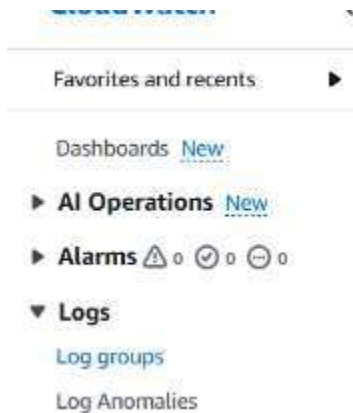


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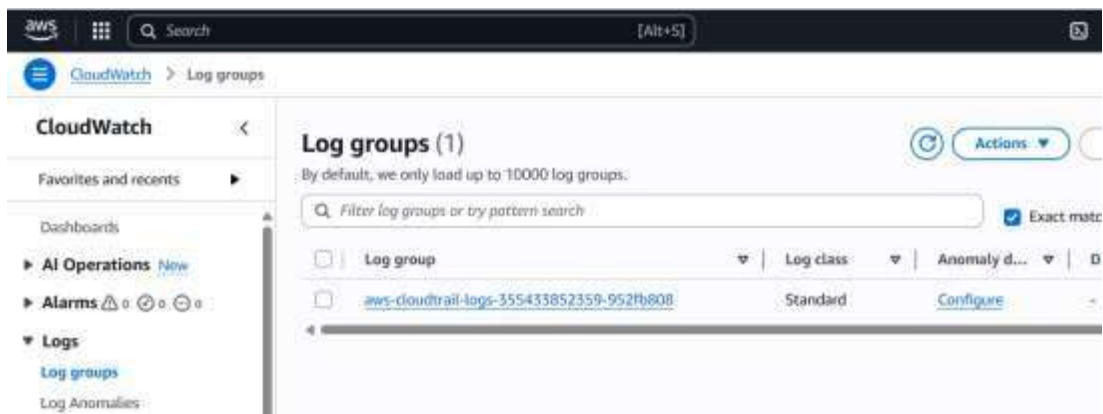
Step 6- Enable CloudWatch Logs, Create new Log Group: /aws/cloudtrail/logs, Allow to create IAM role automatically, then Click Create trail.

Task 2- Configure alerts for suspicious activity.

Step 1-Go to CloudWatch , Log Groups

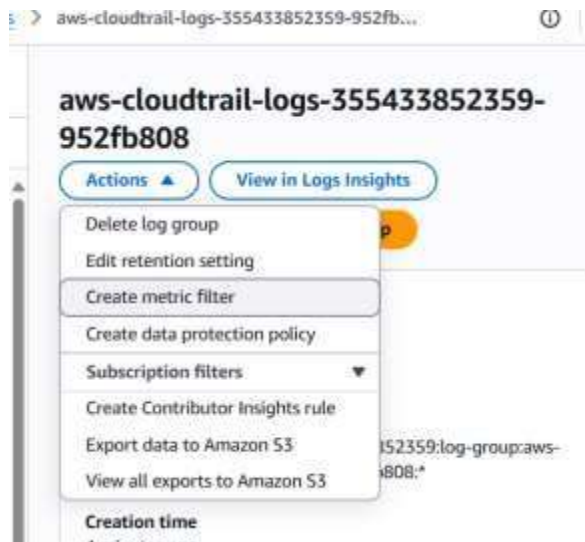


Step 2- Select your log group: /aws/cloudtrail/logs

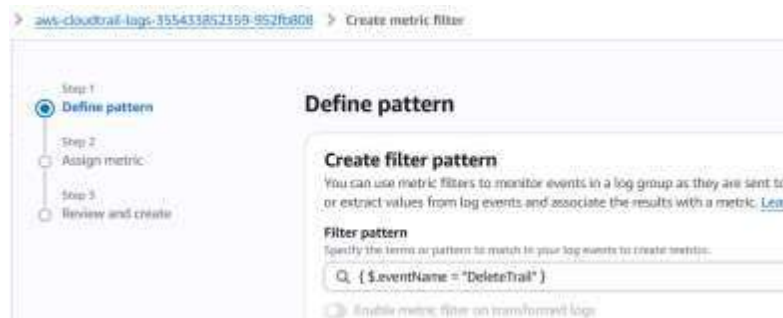


Step 3- Click Actions → Create Metric Filter

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Step 4- Paste filter pattern, and then click next.



Step 5- next step is create filter name, enter the filter name.



Step 6- Then Enter the Matrix Details.

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Metric details

Metric namespace
Namespaces let you group similar metrics. [Learn more](#)

CloudTrailMonitoring [Create new](#)

Namespaces can be up to 255 characters long; all characters are valid except for colon(:) at the start of the name.

Metric name
Metric name identifies this metric, and must be unique within the namespace. [Learn more](#)

DeleteTrailAttempts

Metric name can be up to 255 characters long; all characters are valid except for colon(:), asterisk(*), dollar(\$), and space().

Metric value
Metric value is the value published to the metric name when a Filter Pattern match occurs.

1

Valid metric values are: floating point number (1, 99.3, etc.), numeric field identifiers (\$1, \$2, etc.), or named field identifiers (e.g. \$requestSize for delimited filter pattern or \$.status for JSON-based filter pattern - dollar (\$) or dollar dot (\$) followed by alphanumeric and/or underscore (_) characters).

Default value - optional
The default value is published to the metric when the pattern does not match. If you leave this blank, no value is published when there is no match. [Learn more](#)

0

Unit - optional

Count

Activate Windows
Go to Settings to activate Windows.

Step 7- Then click Create Matrix.

Step 2: Metric [Edit](#)

Assign metric

Filter name DeleteTrailFilter	Metric name DeleteTrailAttempts
Metric namespace CloudTrailMonitoring	Applied on transformed logs -
Metric value 1	Default value 0
Unit Count	

[Cancel](#) [Previous](#) [Create metric filter](#)

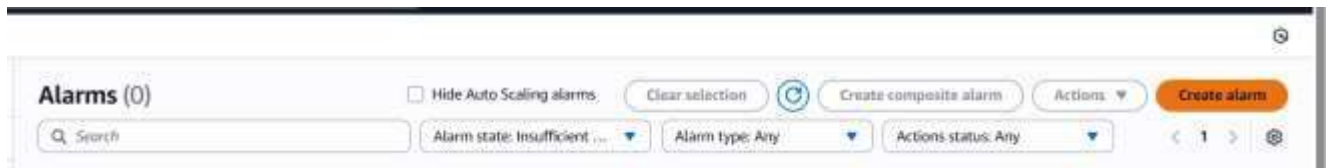
Go to Settings to activate Windows.

Step 8- Matrix filter Created successfully.

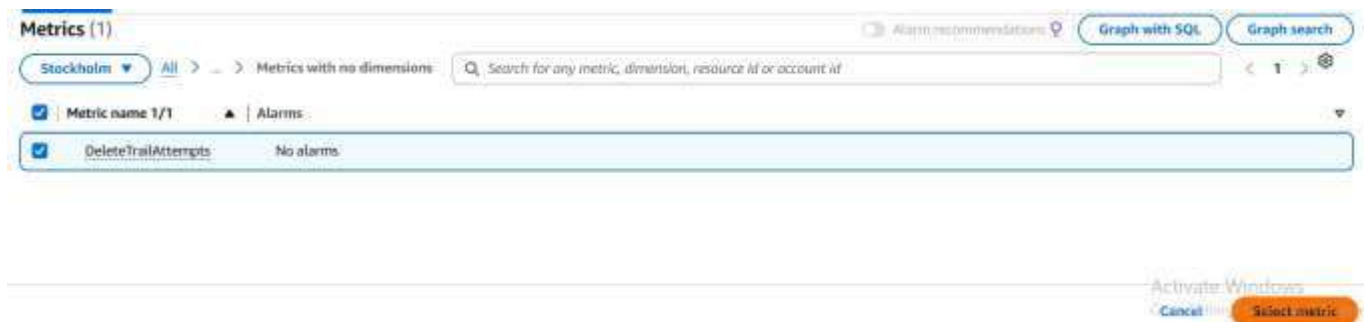


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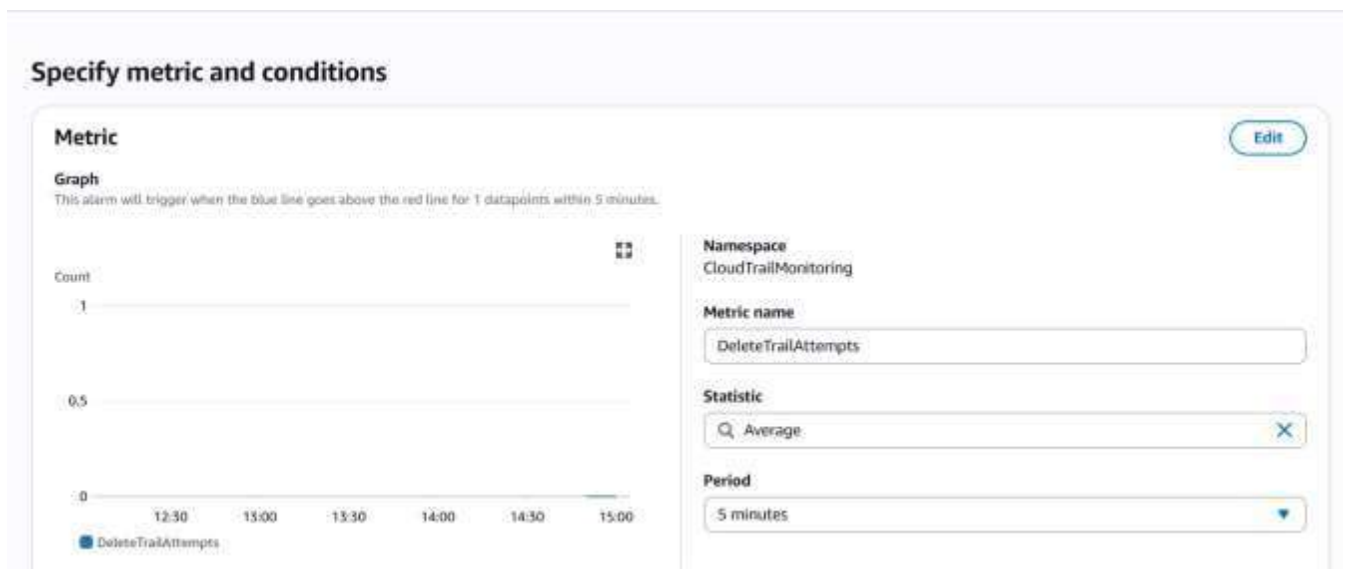
Step 9- Go to CloudWatch → Alarms → Create Alarm



Step 10- Select metric.



Step 11- Set threshold: Static, Condition: Greater than or equal to 1, Period: 5 minute.



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Step 12- Set Condition, and then click Next.

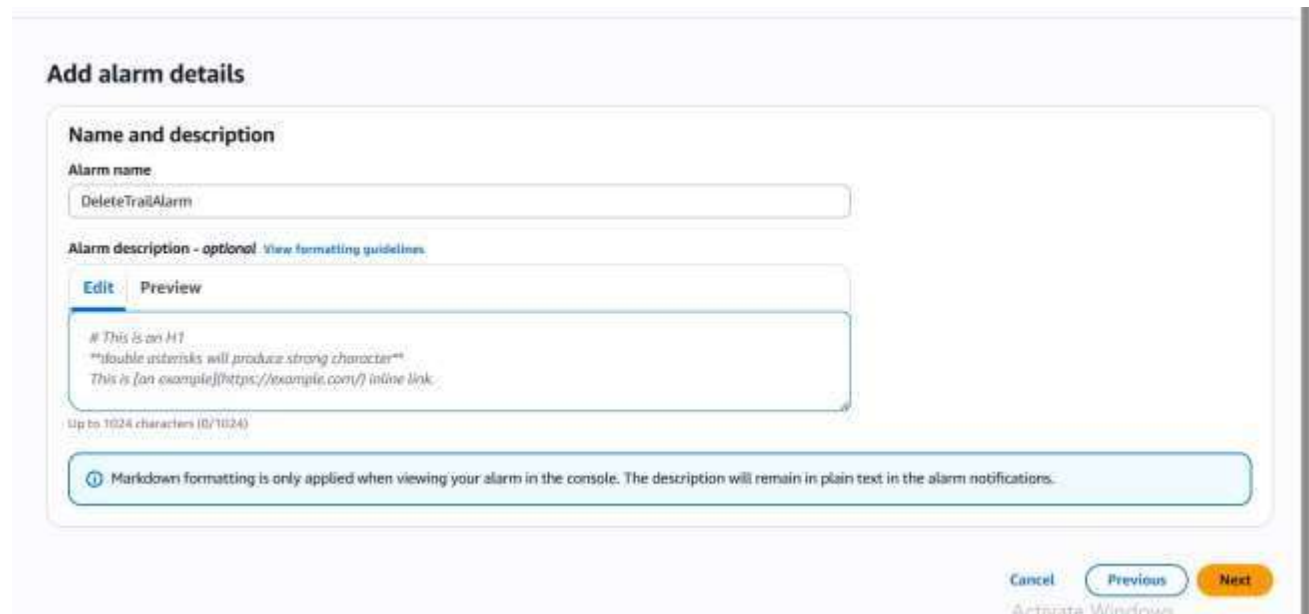
The screenshot shows the 'Conditions' section of the AWS CloudWatch console. Under 'Threshold type', the 'Static' option is selected with the subtext 'Use a value as a threshold'. The 'Whenever DeleteTrailAttempts is...' section shows the 'Greater/Equal' comparison operator selected, with the subtext '>= threshold'. Below this, a text input field contains the value '1'. At the bottom right, there are 'Cancel' and 'Next' buttons. An 'Activate Windows' watermark is visible in the bottom right corner.

Step 13- Add Configure Action Details.

The screenshot shows the 'Configure actions' section of the AWS CloudWatch console. Under 'Alarm state trigger', the 'In alarm' option is selected, with the subtext 'The metric or expression is outside of the defined threshold.' Below this, the 'Send a notification to the following SNS topic' section shows 'Create new topic' selected. A text input field for the topic name contains 'Default_CloudWatch_Alarms_Topic'. Below this, the 'Email endpoints that will receive the notification...' section shows a text input field containing 'nikamsamruddhi2006@gmail.com'. At the bottom right, there is a 'Remove' button and an 'Activate Windows' watermark.

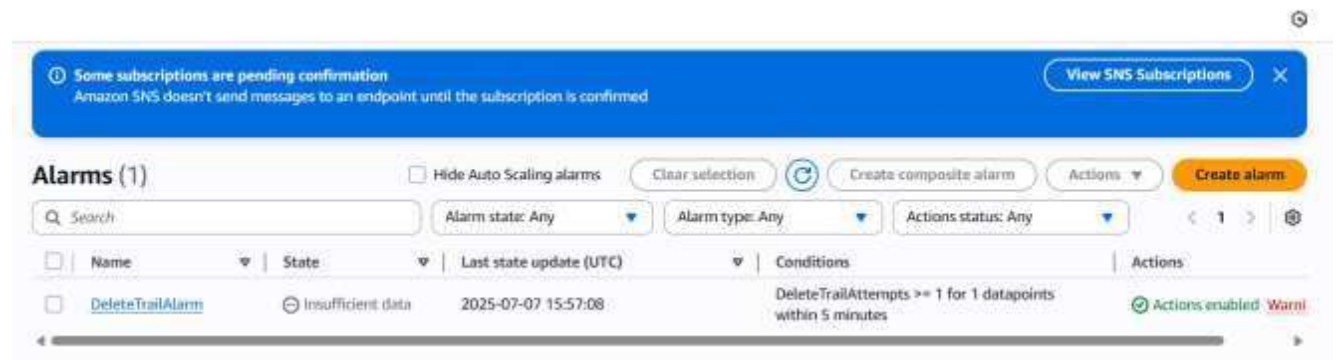
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Step 14- Add Alarm Details, and then click next.



The screenshot shows the 'Add alarm details' form in the AWS CloudWatch console. The form is titled 'Add alarm details' and contains a section for 'Name and description'. Under 'Name and description', there is a text input field for 'Alarm name' containing 'DeleteTrailAlarm'. Below this is a text area for 'Alarm description - optional' with a 'View formatting guidelines' link. The description area has 'Edit' and 'Preview' tabs, with 'Edit' currently selected. The description text is: '# This is an H1', '**double asterisks will produce strong character**', and 'This is [an example](https://example.com/) inline link'. A character count shows 'Up to 1024 characters (0/1024)'. A light blue box contains a note: 'Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.' At the bottom right are 'Cancel', 'Previous', and 'Next' buttons. An 'Activate Windows' watermark is visible at the bottom right.

Step 15- Set Alarm Successfully.



The screenshot shows the AWS Alarms console. At the top, a blue banner states 'Some subscriptions are pending confirmation' with a 'View SNS Subscriptions' button. Below this, the 'Alarms (1)' section is visible. It includes a search bar, filters for 'Alarm state: Any', 'Alarm type: Any', and 'Actions status: Any', and buttons for 'Clear selection', 'Create composite alarm', and 'Create alarm'. A table lists the alarm details:

Name	State	Last state update (UTC)	Conditions	Actions
DeleteTrailAlarm	Insufficient data	2025-07-07 15:57:08	DeleteTrailAttempts >= 1 for 1 datapoints within 5 minutes	Actions enabled Warn